

# Unit 3 :: Complex Analysis

☞ PHY0400404: Mathematical Physics ☞

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# Syllabus

## Unit 3: Complex analysis

Functions of complex variables, analyticity and Cauchy-Riemann conditions, examples of analytic functions; singular functions: poles and branch points, order of singularity; integration of functions with complex variable, Cauchy's integral theorem and Cauchy's integral formula, simply and multiply connected regions; Laurent and Taylor's series expansions, residue theorem with application.

## Complex trigonometric and hyperbolic identities

$$1. \cosh x = \frac{e^x + e^{-x}}{2}$$

$$2. \sinh x = \frac{e^x - e^{-x}}{2}$$

$$3. \cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$4. \sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

Using the above identities, it can be shown that

$$1. \cosh ix = \cos x$$

$$2. \cos ix = \cosh x$$

$$3. \sinh ix = i \sin x$$

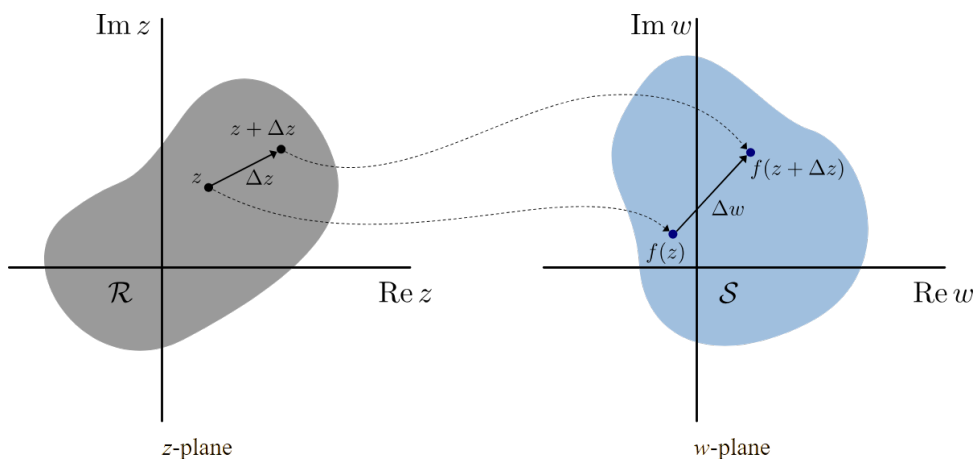
$$4. \sin ix = i \sinh x$$

## 1 Functions of complex variables

Suppose, to each value that a complex variable  $z$  can assume, there corresponds one or more values of a complex variable  $w$ . We then say that  $w$  is a function of  $z$  and write  $w = f(z)$ .

- The complex variable  $z$  is called an *independent variable*
- The complex variable  $w$  is called a *dependent variable*.

### 1.1 Complex derivatives



If a complex function  $f(z)$  is single-valued in some region  $\mathcal{R}$  of the  $z$  plane, the derivative of  $f(z)$  is defined as

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

provided that the limit exists independent of the manner in which  $\Delta z$  approaches 0. In such a case, we say that  $f(z)$  is *differentiable* at  $z$  and  $f'(z)$  is known as the *derivative* of the function  $f(z)$ .

## 1.2 Complex analytic functions

If the derivative  $f'(z)$  exists at all points  $z$  of a region  $\mathcal{R}$ , then  $f(z)$  is said to be analytic in  $\mathcal{R}$  and is referred to as an *analytic function* in  $\mathcal{R}$ .

A function  $f(z)$  is said to be analytic at a point  $z_0$  if there exists a neighborhood  $|z - z_0| < \delta$  at all points of which  $f'(z)$  exists.

**Note:** Analytic functions are sometimes also known as *regular* or *holomorphic* or *monogenic* functions.

## 1.3 Cauchy-Riemann conditions

Given a complex function which can be written as follows:

$$w = f(z) = u(x, y) + iv(x, y)$$

and is analytic over a region  $\mathcal{R}$  on the  $z$ -plane, then the 2D functions  $u(x, y)$  and  $v(x, y)$  satisfy the following Cauchy-Riemann conditions in  $\mathcal{R}$ :

$$\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}}$$

If the partial derivatives of  $u$  and  $v$  are continuous in  $\mathcal{R}$ , then the Cauchy-Riemann equations are sufficient conditions that  $f(z)$  be analytic in  $\mathcal{R}$ .

**Note:** The functions  $u(x, y)$  and  $v(x, y)$  are known as *conjugate functions*.

### Theorem 1

The necessary conditions for a function  $f(z) = u(x, y) + iv(x, y)$  to be analytic at all points in a region  $\mathcal{R}$  are

$$\text{a) } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\text{b) } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

provided that  $\frac{\partial u}{\partial x}$ ,  $\frac{\partial u}{\partial y}$ ,  $\frac{\partial v}{\partial x}$  and  $\frac{\partial v}{\partial y}$  exist.

### Theorem 2

The sufficient conditions for a function  $f(z) = u(x, y) + iv(x, y)$  to be analytic at all points in a region  $\mathcal{R}$  are

- a)  $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$  and  $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$
- b)  $\frac{\partial u}{\partial x}$ ,  $\frac{\partial u}{\partial y}$ ,  $\frac{\partial v}{\partial x}$  and  $\frac{\partial v}{\partial y}$  are continuous functions of  $x$  and  $y$  in the region  $\mathcal{R}$ .

### Theorem 3

Given a function  $f(z) = u(x, y) + iv(x, y)$ , which is analytic over a region  $\mathcal{R}$ , its derivative is obtained as

$$f'(z) = \frac{df}{dz} = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i\frac{\partial u}{\partial y}$$

## 1.4 Examples of analyticity of complex functions

1. Determine whether  $f(z) = \frac{1}{z}$  is analytic or not.

**Solution:** Let us consider

$$\begin{aligned} f(z) &= \frac{1}{z} \\ &= \frac{1}{x + iy} \\ &= \left( \frac{1}{x + iy} \right) \times \left( \frac{x - iy}{x - iy} \right) \\ &= \frac{x - iy}{(x + iy)(x - iy)} \\ &= \frac{x - iy}{x^2 + y^2} \\ \Rightarrow f(z) &= \left( \frac{x}{x^2 + y^2} \right) + i \left( \frac{-y}{x^2 + y^2} \right) \\ &= u(x, y) + iv(x, y) \end{aligned}$$

This gives us

$$u(x, y) = \frac{x}{x^2 + y^2} \quad \text{and} \quad v(x, y) = \frac{-y}{x^2 + y^2}$$

We obtain the partial derivatives as

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{(x^2 + y^2) \times 1 - x \times 2x}{(x^2 + y^2)^2} & \frac{\partial u}{\partial y} &= \frac{(x^2 + y^2) \times 0 - x \times 2y}{(x^2 + y^2)^2} \\ &= \frac{x^2 + y^2 - 2x^2}{(x^2 + y^2)^2} & &= \frac{-2xy}{(x^2 + y^2)^2} \\ &= \frac{y^2 - x^2}{(x^2 + y^2)^2} \end{aligned}$$

$$\begin{aligned}
\frac{\partial v}{\partial x} &= \frac{(x^2 + y^2) \times 0 - (-y) \times 2x}{(x^2 + y^2)^2} & \frac{\partial v}{\partial y} &= \frac{(x^2 + y^2) \times (-1) - (-y) \times 2y}{(x^2 + y^2)^2} \\
&= \frac{2xy}{(x^2 + y^2)^2} & &= \frac{2y^2 - x^2 - y^2}{(x^2 + y^2)^2} \\
& & &= \frac{y^2 - x^2}{(x^2 + y^2)^2}
\end{aligned}$$

From the above partial derivatives, we find that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Hence the Cauchy-Riemann conditions are satisfied. Also the partial derivatives  $\frac{\partial u}{\partial x}$ ,  $\frac{\partial u}{\partial y}$ ,  $\frac{\partial v}{\partial x}$  and  $\frac{\partial v}{\partial y}$  are continuous everywhere except at  $(0, 0)$ .

Therefore the complex function  $f(z) = \frac{1}{z}$  is analytic everywhere except at  $z = 0$ .

2. Show that the function  $f(z) = e^x(\cos y + i \sin y)$  is analytic, and then find its derivative.

**Solution:** Let us consider

$$\begin{aligned}
f(z) &= e^x(\cos y + i \sin y) \\
\Rightarrow f(z) &= (e^x \cos y) + i(e^x \sin y) \\
&= u(x, y) + iv(x, y)
\end{aligned}$$

This gives us

$$u(x, y) = e^x \cos y \quad \text{and} \quad v(x, y) = e^x \sin y$$

We obtain the partial derivatives as

$$\begin{aligned}
\frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}(e^x \cos y) & \frac{\partial u}{\partial y} &= \frac{\partial}{\partial y}(e^x \cos y) \\
&= e^x \cos y & &= -e^x \sin y \\
\\
\frac{\partial v}{\partial x} &= \frac{\partial}{\partial x}(e^x \sin y) & \frac{\partial v}{\partial y} &= \frac{\partial}{\partial y}(e^x \sin y) \\
&= e^x \sin y & &= e^x \cos y
\end{aligned}$$

From the above partial derivatives, we find that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Hence the Cauchy-Riemann conditions are satisfied. Also the partial derivatives  $\frac{\partial u}{\partial x}$ ,  $\frac{\partial u}{\partial y}$ ,  $\frac{\partial v}{\partial x}$  and  $\frac{\partial v}{\partial y}$  are continuous everywhere.

Therefore the complex function  $f(z) = e^x(\cos y + i \sin y)$  is analytic everywhere.

Finally, the derivative of the given function is obtained as

$$\begin{aligned} f'(z) &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \\ &= e^x \cos y + i e^x \sin y \\ \implies f'(z) &= e^x(\cos y + i \sin y) \end{aligned}$$

3. Test the analyticity of the function  $w = \sin z$  and thereby show that  $\frac{d}{dz}(\sin z) = \cos z$ .

**Solution:** We have

$$\begin{aligned} w &= \sin z \\ &= \sin(x + iy) \\ &= \sin x \cos iy + \cos x \sin iy \\ &= \sin x(\cosh y) + \cos x(i \sinh y) \\ \implies w &= f(z) = \sin x \cosh y + i \cos x \sinh y \end{aligned}$$

This gives us

$$u(x, y) = \sin x \cosh y \quad \text{and} \quad v(x, y) = \cos x \sinh y$$

The partial derivatives are

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}(\sin x \cosh y) & \frac{\partial u}{\partial y} &= \frac{\partial}{\partial y}(\sin x \cosh y) \\ &= \cos x \cosh y & &= \sin x \sinh y \\ \frac{\partial v}{\partial x} &= \frac{\partial}{\partial x}(\cos x \sinh y) & \frac{\partial v}{\partial y} &= \frac{\partial}{\partial y}(\cos x \sinh y) \\ &= -\sin x \sinh y & &= \cos x \cosh y \end{aligned}$$

From the above partial derivatives, we find that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Hence the Cauchy-Riemann conditions are satisfied. Also the partial derivatives  $\frac{\partial u}{\partial x}$ ,  $\frac{\partial u}{\partial y}$ ,  $\frac{\partial v}{\partial x}$  and  $\frac{\partial v}{\partial y}$  are continuous everywhere.

Therefore the complex function  $f(z) = \sin z$  is analytic everywhere.

Now, we evaluate the derivative as follows

$$\begin{aligned} \frac{d}{dz}(\sin z) &= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \\ &= \cos x \cosh y + i(-\sin x \sinh y) \\ &= \cos x \cosh y - \sin x(i \sinh y) \\ &= \cos x(\cos iy) - \sin x(\sin iy) \\ &= \cos(x + iy) \\ \implies \frac{d}{dz}(\sin z) &= \cos z \end{aligned}$$



## **2 Singularities of complex functions**

### **2.1 Poles**

### **2.2 Branch points**

### **3 Complex integration**

#### **3.1 Simply and multiply connected regions**

#### **3.2 Cauchy's integral theorem**

#### **3.3 Cauchy's integral formula**

## **4 Complex series**

### **4.1 Taylor's series expansion of a complex function**

### **4.2 Laurent's series expansion of a complex function**

#### **4.2.1 Residue theorem**